Risks in mining

Demand and Supply

- **Demand and supply analysis** is the study of how buyers and sellers interact to determine transaction prices and quantities.
- **Demand**, in economics, is the willingness and ability of consumers to purchase a given amount of a good or service at a given price.
- **Supply** is the willingness of sellers to offer a given quantity of a good or service for a given price.
- The demand and supply model is useful in explaining how price and quantity traded are determined and how external influences affect the values of those variables. Buyers' behavior is captured in the demand function and its graphical equivalent, the demand curve. This curve shows both the highest price buyers are willing to pay for each quantity, and the highest quantity buyers are willing and able to purchase at each price. Sellers' behavior is captured in the supply function and its graphical equivalent, the supply curve. This curve shows simultaneously the lowest price sellers are willing to accept for each quantity and the highest quantity sellers are willing to offer at each price.

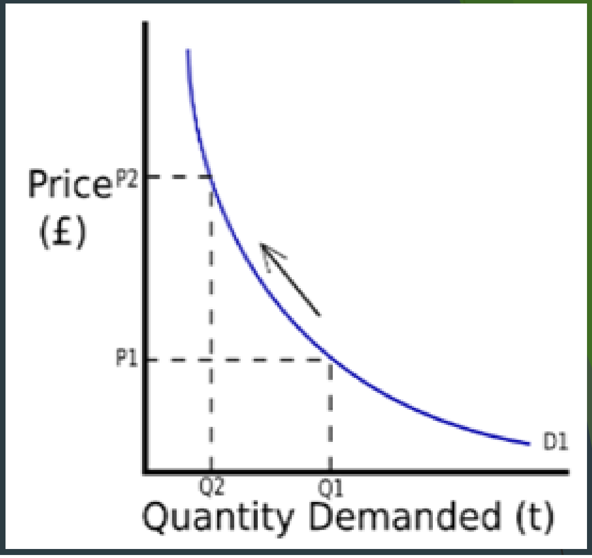
The Demand Function and the Demand Curve

- The quantity consumers are willing to buy clearly depends on a number of different factors called variables, Perhaps the most important of those variables is the item's own price. In general, economists believe that as the price of a good rises, buyers will choose to buy less of it, and as its price falls, they buy more.
- Although a good's own price is important in determining consumers' willingness to purchase it, other variables also have influence on that decision, such as consumers' incomes, their tastes and preferences, the prices of other goods that serve as substitutes or complements, and so on. Economists attempt to capture all of these influences in a relationship called the **demand function**. (In general, a function is a relationship that assigns a unique value to a dependent variable for any given set of values of a group of independent variables.) We represent such a demand function in Equation 1:

$$Q_x^d = f(P_x, I, P_y, \ldots) \qquad (1)$$

where Q^d_x represents the quantity demanded of some good *X*, P_x is the price per unit of good *X*, *I* is consumers' income, and P_y is the price of another good, *Y*

- In economics, a demand curve is a graph depicting the relationship between the price of a certain commodity and the quantity of that commodity that is demanded at that price. Demand curves can be used either for the price-quantity relationship for an individual consumer, or for all consumers in a particular market



Demand Function

Algebraically, the demand function can be represented as:

Qd = *f* (P, Ps, Pc, Y, A, A', N, CP, PE, TA, T/S*)*

- where Qd = quantity demanded of the good or service
- P = price of the good or service
- Ps = price of substitute goods or services
- Pc = price of complementary goods or services.
- Y = income of consumers.
- A = advertising expenditures (and other marketing expenditures)
- A' = competitors' advertising expenditures on the good or service.
- N = population (and other demographic factors).
- CP = consumer tastes and preferences for the good or service.
- PE= expected (future) changes in price.
- TA= adjustment time period.
- T/S = taxes or subsidies.

Demand and Supply as Equations

- Let's now look at demand and supply as equations. Here is a demand equation: ***Qd = 1,500 – 32P***
- To see what this equation says, we let price ***(P)*** *in the equation equal $10 and then solve* for quantity demanded ***Qd****. We get* Q_d ***= 1,180.***

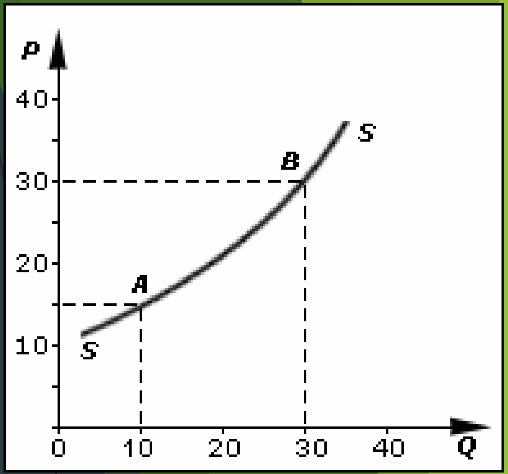
$$Q_d = 1{,}500 - 32(10) = 1{,}180$$

- So this equation says that if price is $10, it follows that quantity demanded is 1,180 units.
- We could find other quantities demanded by plugging in different dollar amounts for price (*P).*

$$Q_x^d = 8.4 - 0.4P_x + 0.06I - 0.01P_y \qquad (2)$$

The Supply Function and the Supply Curve

- The willingness and ability to sell a good or service is called supply. *In general, producers are willing to sell their product for a price as long as that price is at least as high as the cost to produce an additional unit of the product. It follows that the willingness to supply, called the* ***supply function, depends on the price at which the good can be sold as well as the cost of production for an additional unit of the good****.* ***The greater the difference between those two values, the greater is the willingness of producers to supply the good.***



For simplicity, we can assume that the only input in a production process is labor that must be purchased in the labor market. The price of an hour of labor is the wage rate, or *W. Hence, we can say that (for any given level of technology) the willingness to supply a good depends on the price of that good and the wage rate. This concept is captured in the following equation, which represents an individual seller's supply function:*

$$Q_x^s = f(P_x, W, \ldots)$$

SUPPLY FUNCTION

Supply function refers to the mathematical expression of the quantity supplied and the factors that determine the supply.

$$S_x = f(P_x\ P_1 \ldots P_n\ C, T, G, O)$$

Where S_x refers to the quantity supplied of good X

P_x refers to the price of good X

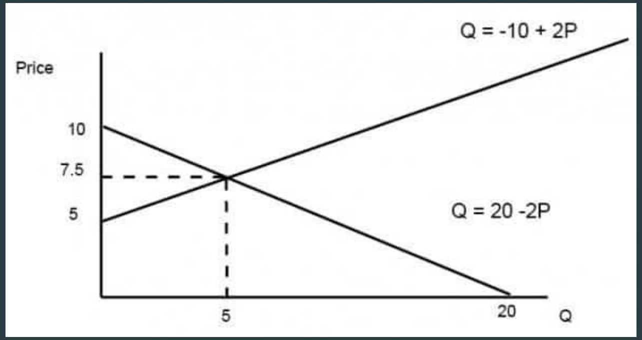
$P_1 \ldots P_n$ refers to the price of related goods

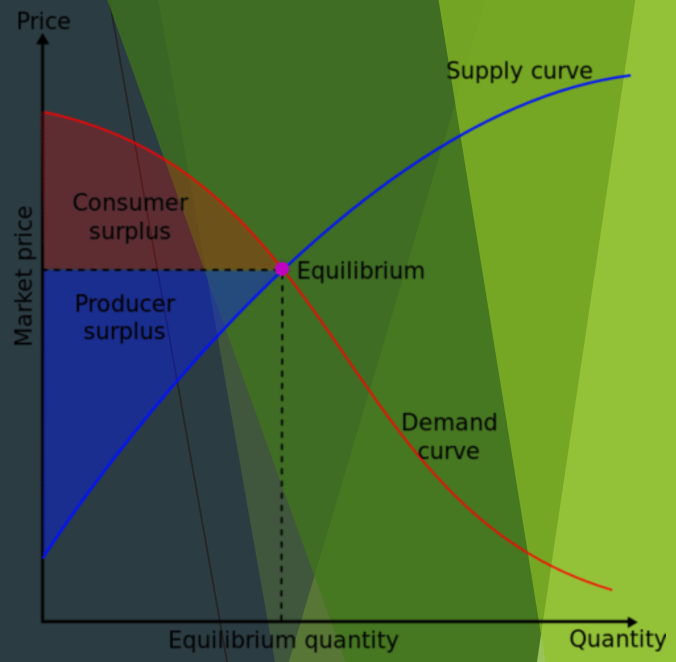
C refers to the cost of production

T refers to the state of technology.

G refers to the goal of the firm

O refers to other factors.





Elasticity of Demand

- **Elasticity of demand is an important variation on the concept of demand. Demand can be classified as elastic, inelastic or unitary.**
- An **elastic demand is one in which the change in quantity demanded due to a change in price is large. An inelastic demand is one in which the change in quantity demanded due to a change in price is small**
- The formula used here for computing elasticity of demand is:

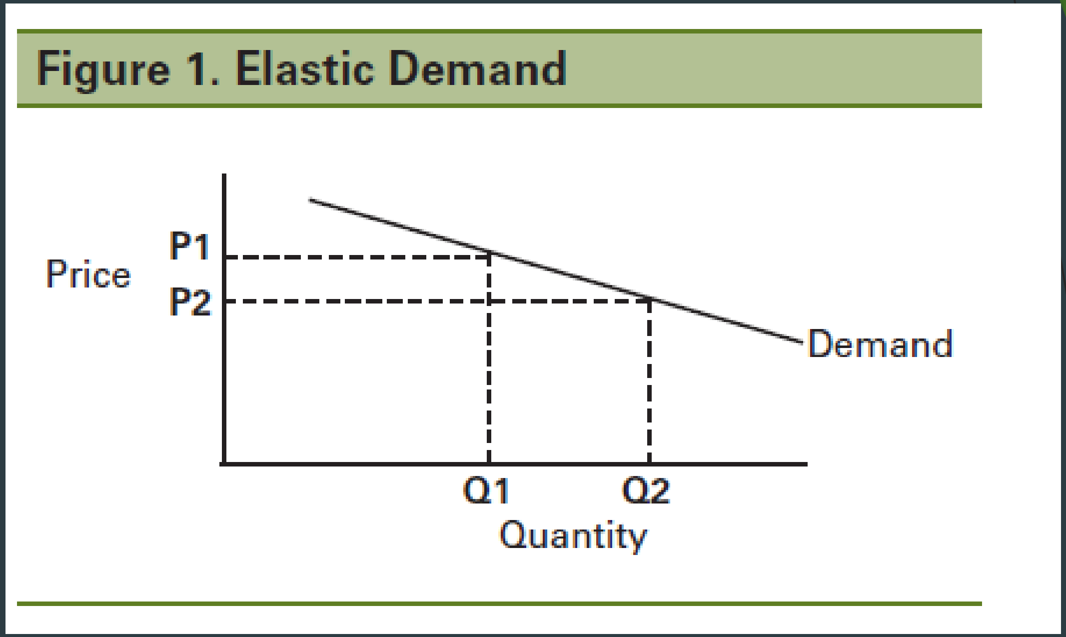
  <u>(Q1 – Q2) / (Q1 + Q2)</u>

  (P1 – P2) / (P1 + P2)

$$E_S = \frac{\%\Delta Q}{\%\Delta P}$$

- If the formula creates an *absolute value greater than 1, the demand is elastic. In other words, quantity changes faster than price. If the value is less than 1, demand is inelastic. In other words, quantity changes slower than price. If the number is equal to 1, elasticity of demand is unitary. In other words, quantity changes at the same rate as price.*

Elastic Demand

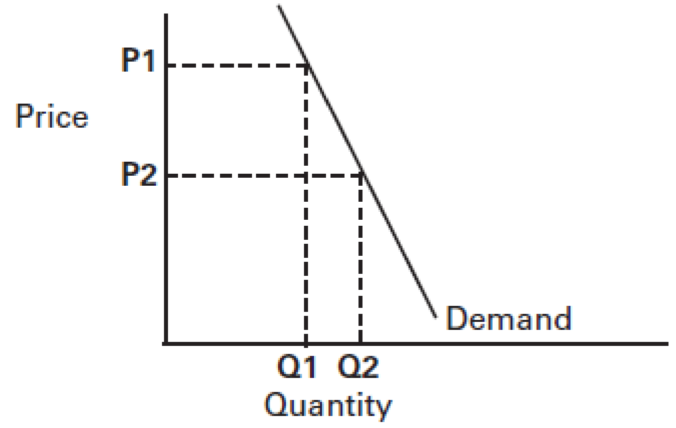


- When the price decreases from $10 per unit to $8 per unit, the quantity sold increases from 30 units to 50 units. The elasticity coefficient is 2.25.

Inelastic Demand

- Inelastic demand is shown in Figure 2. Note that a change in price results in only a *small change in quantity demanded. In other words, the quantity demanded is not very responsive to changes in price. Examples of this are necessities like food and fuel. Consumers will not reduce their food purchases if food prices rise, although there may be shifts in the types of food they purchase. Also, consumers will not greatly change their driving behavior if gasoline prices rise.*

Figure 2. Inelastic Demand



When the price decreases from $12 to $6 (50%), the quantity of demand increases from 40 to only 50 (25%). The elasticity coefficient is .33

Unitary Elasticity

- If the elasticity coefficient is equal to one, demand is unitarily elastic as shown in Figure 3. For example, a 10% quantity change divided by a 10% price change is one. This means that a 1% change in quantity occurs for every 1% change in price.

Figure 3. Unitary Elasticity